

**Equipment Procedure  
for Determination of Viscosity and Density Using Anton Paar SVM 3001**

**Equipment Number: EQ-01 & EQ-02  
Serial Number: 99111434 & 84027702**



## 1. PURPOSE

This standard operating procedure is intended to provide users the guide on the operation and maintenance of Anton Paar SVM 3001.

## 2. SCOPE

This standard operating procedure is employed in GCC Oil Lab primarily for performing Dynamic, Kinematic viscosity and Density tests with Anton Paar SVM 3001 using Stabinger method according to the ASTM D7042 test method and related test specifications.

## 3. TERMS AND DEFINITIONS

3.1.1 None

## 4. RESPONSIBILITIES

4.1 Use, care and maintenance of the Anton Paar SVM 3001 is directly assigned to the end user who is usually the Sr. laboratory chemist. This responsibility may be shared to other qualified user(s) upon the consent of the Sr. laboratory chemist.

4.2 Application of this responsibility is based on the Memorandum Receipt and further supported by a Memorandum of Authorization both signed by the General Manager.

## 5. SAFETY PRECAUTIONS

5.1 Use a proper electrical outlet of correct voltage and current handling capacity.

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| Doc.: EOP [6]-04-K              | Version No.00                     | Issue Date: 1/11/2023           | Revision No: 00 | Date:1/11/2023 |
| Prepared by:<br>Quality Manager | Reviewed by:<br>Technical Manager | Approved by:<br>General Manager | Page 1 of 5     |                |

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| <b>GCC Lab</b><br>المفتبر الفليبي | Gulf Laboratory Technical Services                                                                        |
|                                   | <b>Equipment Procedure</b><br><b>for Determination of Viscosity and Density Using Anton Paar SVM 3001</b> |

5.2 Disconnect from the power supply prior to maintenance and servicing.

5.3 Ensure that the operating switch operates properly.

## 6. PRINCIPLES OF OPERATION

### 6.1 Unit Design

This equipment is specifically designed for use in accordance with the applicable international standard test methods. It covers a measurement temperature range of -20 to 135 °C. Can be used only for the following liquids:

1. Standard Base Oils.
2. Lubricant Oils.
3. Insulating Oils.
4. Hydraulic Liquids.
5. Fuels.
6. Pure Hydrocarbons.

### 7. Sample Preparation

7.1 Homogenize the sample in the original container by gently shaking or stirring. Use a laboratory stirrer with magnetic rods at low speed to avoid bubble generation. Pouring sample into a separate container will change the sample's properties as contaminants will settle on the bottom of the vessel.

7.2 Remove rough contaminants and swarfs using a sieve with appropriate mesh width or a filter with appropriate aperture. - For heated samples heat also the screen. - Remove only particles which are bigger than the maximum allowed size. Removing more particles than necessary for measurement would falsify the results.

7.3 Remove ferromagnetic particles either using a laboratory stirrer with magnetic rod (for occasional measurements and to remove large swarfs) or use the optionally available Magnetic Particle Trap.

7.4 Water content can be removed by pouring the sample through a filter paper. The paper will absorb the water, but it will filter particle content too, which may falsify the measuring results.

7.5 Recap the container tightly and shake it thoroughly for one minute to complete mixing.

7.6 Leave the container in the oven for another 10 minutes to allow bubbles to escape from the sample liquid.

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| Prepared by:<br>Quality Manager | Reviewed by:<br>Technical Manager | Approved by:<br>General Manager | Page 2 of 5     |                |

**Equipment Procedure  
for Determination of Viscosity and Density Using Anton Paar SVM 3001**

7.7 Suck the sample into the syringe and measure it immediately.

7.8 Keep the sample at temperature until measurements are completed. This avoids separation within the sample when cooling down.

If you cannot measured the sample after the above preparation routine (e.g. due to gassing at measuring temperature), perform the following extended preparation steps:

7.9 Fill enough sample for your measurements into a 100 m glass flask and loosely stopper it.

7.10 Put the flask into a bath with boiling water or into a laboratory oven set to 100 °C for 30 minutes

7.11 Remove the flask from the water bath or from the oven and stopper it tightly.

7.12 Use paper tissue and cover the area of stopper and neck of the flask. This measure will help to avoid possible escaping hot liquid running over your hands when opening the flask later.

7.13 Shake the flask for 60 seconds.

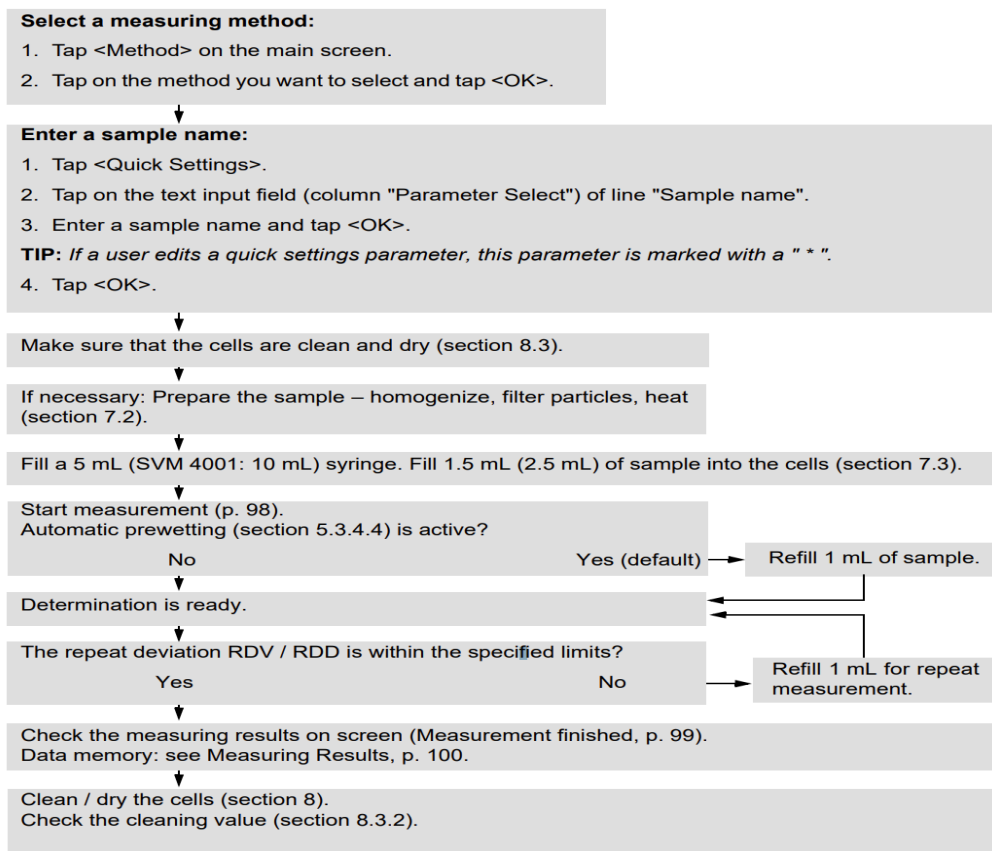
7.14 Open the flask very slowly to release the inside pressure. Be cautious, hot sample liquid can get out during this procedure.

7.15 Fill the test specimen into the syringe and measure the sample immediately.

**8. OPERATIONAL DETAILS According to Manual**

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**Equipment Procedure  
for Determination of Viscosity and Density Using Anton Paar SVM 3001**



## 9. Calibration and Verification

9.1 Refer to manufacturer manual.

9.2 Calibration of with Certified Viscosity Reference Standards once a month.

## 10. Maintenance

10.1 Remove the sample from the cells: Push it out of the cells with the empty syringe or suck the sample back into the syringe. Remove low viscosity samples using the built-in air pump.

10.2 Fill a syringe with approx. 1.5 mL (SVM 4001: 2.5 mL) of solvent and fill the solvent into the measuring cells.

10.3 Start the motor to improve the cleaning performance in the viscosity cell only and stop it again.

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**Equipment Procedure  
for Determination of Viscosity and Density Using Anton Paar SVM 3001**

- 10.4 Move the plunger of the syringe vigorously forth and back when the motor is at filling speed; this way you improve the cleaning performance both in the density oscillator and in the viscosity cell. Note: The number of forth-back movements needs to be increased with increasing viscosity respectively contamination of the oil.
- 10.5 When filling from the side: To avoid spillage, suck back the liquid from the cells into the syringe.
- 10.6 Use the air pump to remove the sample-solvent mixture from the cells.
- 10.7 Depending on the sample, repeat this procedure. 1.5 mL (SVM 4001: 2.5 mL) of solvent are sufficient for repeated cleaning cycles.
- 10.8 Perform a final rinse with approx. 2 mL of fresh solvent to remove any residues.
- 10.9 Allow a sufficiently long drying time to be sure that the solvent can dry up completely.

## 10.0 REFERENCE

- 10.1 SVM 3001 Operation and Instruction Manual
- 10.2 Standard Method ASTM D7042

## 12.0 RECORDS

|                          |                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Identification           | Equipment Data Record<br>Equipment Maintenance / Calibration Schedule<br>Equipment Operation Manual<br>Equipment Maintenance Record |
| Collection and Filing    | Filed by the authorized user at the Chemistry Lab                                                                                   |
| Index                    | Equipment Data Record<br>Equipment Maintenance Record                                                                               |
| Access                   | Authorized user                                                                                                                     |
| Storage                  | Office and Laboratory Environments                                                                                                  |
| Maintenance and Disposal | Maintained as active file for three years and disposed after 10 years                                                               |